

Lecture 3: Decoding IP Addresses

Every device on a network needs a unique address to receive data correctly — much like every house needs an address for mail to be delivered without confusion.

The Origin of TCP/IP

1969 → ARPANET created
1974 → TCP/IP protocols developed

When ARPANET connected computers across different countries (e.g. USA and Europe), each region initially followed its own system of communication, causing confusion between nodes.

To fix this, a common set of **standard protocols** was introduced:

***TCP/IP** — a standardized set of rules that every device on the network must follow, so that no matter which country or system a device belongs to, all devices "speak the same language."*

The **TCP/IP suite** determines the **packet size** and structures its contents, including the **source** and **destination IP addresses**.

What is an IP Address?

IP stands for **Internet Protocol**.

*Think of it like the **Post Office System** — to send a letter (packet), you need the address of where to send it.*

An **IP Address** is a **unique address** assigned to every device connected to a network, so that data can be correctly routed to it.

IPv4 (Internet Protocol version 4)

- ♦ IPv4 addresses are **32-bit** in length.
- ♦ 32 bits = **4 bytes** (since 1 byte = 8 bits).
- ♦ These 4 bytes are written as 4 numbers separated by dots (dotted-decimal notation).

32 bits = [8 bits] . [8 bits] . [8 bits] . [8 bits]

Example:

192.36.15.8

Each of the 4 sections can range from:

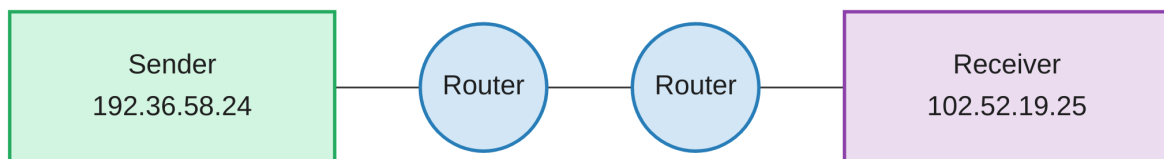
0 - 255

Why 0–255?

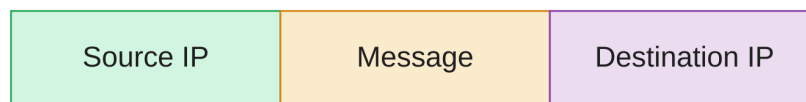
1 byte = 8 bits
 $2^8 = 256$ possible values
→ Range = 0 to 255

- ♦ 0 represents **all bits 0**
- ♦ 255 represents **all bits 1**

How IP Addressing Works



A packet travelling across the network carries this structure:



- ♦ **Routers** act as intermediaries between the sender and receiver and forward the message toward its destination based on the destination IP address.
- ♦ When any device connects to the internet, it is assigned an IP address.

IPv4 Limitation: The Case for IPv6

Total number of unique IPv4 addresses possible:

$2^{32} = 256 \times 256 \times 256 \times 256 \approx 4.3 \text{ billion } (\sim 400 \text{ crore}) \text{ addresses}$

This sounds like a lot, but the number of devices connected to the internet worldwide has already crossed **1000 crore+ (10 billion+)** — far more than IPv4 can support.

To solve this, **IPv6** was introduced:

	IPv4	IPv6
Bits	32-bit	128-bit
Format	Decimal (dotted)	Hexadecimal
Total addresses	$2^{32} \approx 4.3$ billion	2^{128} (extremely large — built to last for the future)

IPv6 addresses are written in **hexadecimal form** to keep the (much longer) address compact and readable.

Who Allocates IP Addresses?

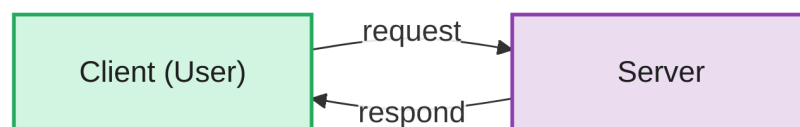
IP addresses are allocated by an **Internet Service Provider (ISP)** — for example, Jio, Airtel, Vi, etc. When a device connects to the internet through an ISP, it is given an IP address to communicate with the rest of the network.

Domain Name System (DNS)

If every website has an IP address, why do we type `google.com` in the browser instead of typing an IP address?

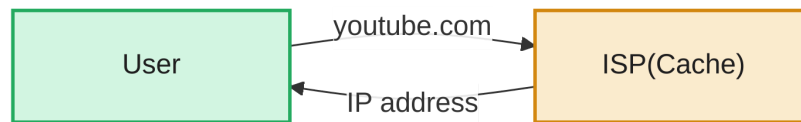
Humans find it far easier to remember **names/words** than long strings of numbers — this is exactly why **DNS** was developed (around **1980**).

DNS (Domain Name System) converts a human-friendly **domain name** into the numeric **IP address** of the server that hosts it.



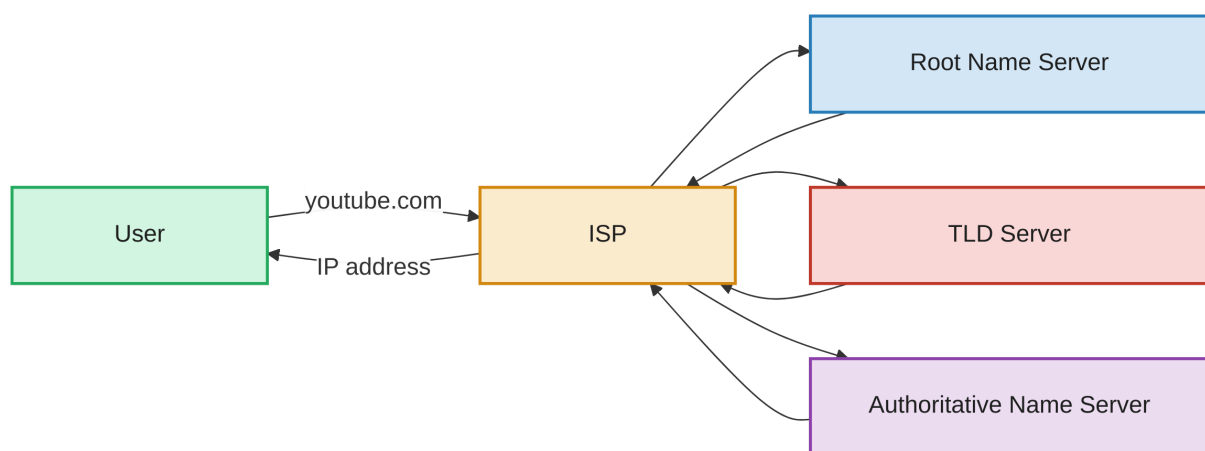
- ♦ **Client** — sends the request
- ♦ **Server** — serves the request of the client (responds with the requested resource/address)

How DNS Resolution Works



- ♦ The ISP maintains a **cache** — a temporary, fast memory that stores IP addresses of **frequently visited websites**.
- ♦ If the IP address is already in the ISP's cache, it is returned to the user immediately.

But what if the ISP does *not* have the requested IP address in its cache?



Step-by-step flow:

1. User sends a request for `youtube.com` , which reaches the **ISP**.
2. The ISP first checks its own **cache** for a previously resolved IP address.
3. If not found in cache, the ISP queries one of the **Root Name Servers**.
4. The Root Name Server does not hold YouTube's IP directly — it only knows about top-level domains (`.com` , `.in` , `.org` , etc.) and directs the ISP to the correct `.com` **TLD Server**.
5. The ISP queries the `.com` TLD Server.
6. The TLD Server does not hold YouTube's IP either — it looks up which **Authoritative Name Server** is responsible for `youtube.com` and returns that server's address to the ISP.
7. The ISP queries that Authoritative Name Server directly.
8. The **Authoritative Name Server** holds the actual DNS records for the domain, including its IP address. It selects an appropriate IP and sends it back to the ISP.

Important notes:

- ♦ The IP address is returned to the **ISP**, not directly to the user — the ISP sends it forward.

- ♦ There are only **13 logical Root Name Servers** in the entire world.
- ♦ This entire multi-step lookup only happens when the ISP's cache misses — because of the large volume of different domain requests it receives, it cannot store everything permanently.
- ♦ The device performing this lookup is technically called the **DNS Resolver**. The ISP is the DNS Resolver by default, but a user can also choose a different one, such as Google's `8.8.8.8` or Cloudflare's `1.1.1.1`.

Static vs Dynamic IP Address

STATIC IP ADDRESS	DYNAMIC IP ADDRESS
A fixed IP address for a device — it does not change	An IP address that changes over time
Used by devices that are 100% online at all times	Used by devices that go offline at times
Example: servers of Amazon, Flipkart	Example: your phone , personal devices

Static IPs make sense for always-on servers so they can always be found at the same address; dynamic IPs are reassigned as needed for devices that connect and disconnect frequently.

Classes in IP Addresses

IPv4 address space is divided into **classes**, based on the range of the first byte (octet). Each class reserves a different share of the total address space.

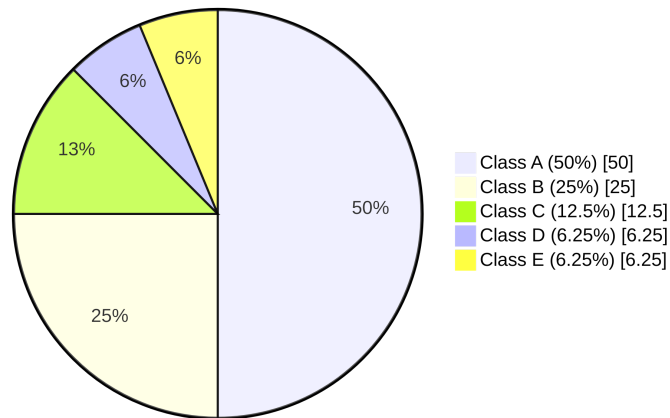
CLASS	SHARE OF IP ADDRESSES	FIRST OCTET RANGE	FORMAT
Class A	50%	0 – 127	<code>0-127 ._.._.</code>
Class B	25%	128 – 191	<code>128-191 ._.._.</code>
Class C	12.5%	192 – 223	<code>192-223 ._.._.</code>
Class D	6.25%	224 – 239	<code>224-239 ._.._.</code>
Class E	6.25%	240 – 255	<code>240-255 ._.._.</code>

How the split is derived:

Class A → $256 / 2 = 128$ → 50% → range 0-127
Class B → remaining half split again → 25% → range 128-191
Class C → further split → 12.5% → range 192-223
Class D → further split → 6.25% → range 224-239
Class E → remaining → 6.25% → range 240-255

Each class split is essentially **halving the remaining address space** repeatedly — Class A takes the first half, Class B takes half of what's left, and so on.

Distribution of IP Address Classes



Summary

- ♦ TCP/IP is the standardized set of protocols that all devices on a network must follow, established after ARPANET (1969) to avoid confusion between differing regional systems; it defines the packet size and what each packet contains, including the source and destination address.
- ♦ An IP Address works like a postal address — it uniquely identifies a device so that data (packets) can be delivered to the correct destination.
- ♦ IPv4 addresses are 32-bit, written as 4 bytes (8 bits each) separated by dots, with each byte ranging from 0–255.
- ♦ Every packet carries a Source IP and a Destination IP, and a router uses these to forward the packet toward its destination.
- ♦ IPv4 can support only about 4.3 billion (2^{32}) addresses, which is insufficient for the 1000 crore+ devices connected worldwide today.
- ♦ IPv6 solves this with 128-bit addresses (2^{128} possible addresses), written in hexadecimal form.
- ♦ IP addresses are allocated to devices by Internet Service Providers (ISPs) such as Jio, Airtel, and Vi.
- ♦ The Domain Name System (DNS), introduced around 1980, resolves a domain name to its corresponding IP address.
- ♦ DNS resolves a domain name to its IP address, through a chain of servers: the ISP (acting as the DNS Resolver) checks its cache first, then asks a Root Name Server, which points to the right TLD Server, which points to the Authoritative Name Server that holds the actual IP.
- ♦ A Static IP Address stays fixed and is used by always-online devices like servers (Amazon, Flipkart), while a Dynamic IP Address changes over time and is used by devices like personal phones that go offline periodically.
- ♦ IP addresses are divided into Classes A–E based on the value of the first octet (out of 256 possible values), with Class A covering 50% of the address space, Class B 25%, Class C 12.5%, and Classes D and E 6.25% each.